

Monocept Consulting — Tech Lead Interview Prep

Candidate: Bhavesh Jakhar | Java Backend Developer / Technical Lead, 5 yrs (PayPal via HCL/DXC)

This guide turns your resume into ready-to-speak answers. Use your own project names (Open Banking, TrackMate) as the spine of every story — interviewers trust specifics far more than textbook definitions.

1. Introduction & Experience

"Tell me about your professional journey, projects, tech stack, and responsibilities"

Structure this as a 90-second narrative, not a list:

"I'm a Java backend developer with 5 years of experience, all within the PayPal ecosystem, progressing from Software Engineer to Lead Engineer to Technical Lead. I started at DXC Technology on PayPal TrackMate, an enterprise privacy compliance platform — I helped build the foundational data model and core scanning/categorization microservices using Spring Boot, Hibernate, and GraphQL, covering PayPal's 8 global brands.

I then moved to HCL Technologies and took over as Lead Engineer on the same TrackMate platform, where I scaled it significantly — architected 30+ GraphQL APIs with multi-tier caching, supporting GDPR/CCPA compliance across 125+ domains, managing 4,465+ web trackers and generating automated compliance reports covering 944K+ pages. I also acted as secondary Scrum Master and mentored 4+ engineers.

Since January 2026, I've moved into a Technical Lead role on a new strategic initiative — PayPal Risk Onboarding's Open Banking Integration. This is a real-time risk decisioning system projected to deliver \$4M in annual loss savings and \$5M in revenue enablement by gating high-risk merchant onboarding on real-time bank confirmation. I'm leading the backend architecture — a Spring Boot microservice that acts as a mapping/decisioning layer between SPI v2 bank-confirmation contracts and merchant onboarding systems, using JSON Logic-based decision graphs to drive dynamic UI behavior, and AMQ events for asynchronous decisioning.

Across all of this my core stack has been Java 8/11/17, Spring Boot, Spring Cloud/Data JPA, Hibernate, REST and GraphQL, Kafka/AMQ for event-driven flows, MySQL/Oracle, AWS, and an observability stack of Grafana, Prometheus, Datadog, and CAL logs. I also actively use AI-augmented engineering tools — Claude, GitHub Copilot, Cursor — to accelerate delivery while maintaining code quality."

"Explain your current project architecture and your role in it"

Walk through Open Banking Phase 0/1 as a system, not a feature:

1. **Trigger:** a high-risk merchant attempts onboarding or product provisioning.

2. **Decision gateway:** your Java/Spring Boot microservice intercepts the request and checks whether SPI v2 bank-confirmation requirements are satisfied.
3. **Mapping layer:** your service translates risk-policy rules into a contract that onboarding systems understand — this is the part you architected.
4. **Dynamic UI:** JSON Logic-based decision graphs determine what contingencies/forms get shown to the merchant during onboarding, based on bank-confirmation status.
5. **Async decisioning:** Financial Instrument events flow over AMQ; your service consumes these asynchronously so provisioning decisions update without blocking the onboarding flow.
6. **Cross-team contracts:** you own the API contract negotiation with Risk Policy, Open Banking, Wallet, and ALP teams — this is the Tech Lead dimension interviewers want to hear (not just code, but interface ownership).
7. **Quality/Ops:** JUnit/Mockito/Postman test layers, Maven artifact publishing for reuse, and Grafana/Datadog/CAL-based P95/P99 monitoring for production triage.

Your role specifically: architecture decisions for the mapping layer, API contract ownership, mentoring, and production reliability — frame yourself as owning *outcomes* (\$4M/\$5M business impact), not just code.

2. Requirement Analysis & Tech Lead Responsibilities

When you receive a requirement, how do you analyze it?

1. **Clarify the business outcome first** — e.g., for Open Banking: "why does gating provisioning on bank confirmation reduce loss?" Understanding the *why* prevents over/under-engineering.
2. **Identify actors and systems involved** — which upstream/downstream teams (Risk Policy, Wallet, ALP) does this touch?
3. **List data contracts** — what data is available today (e.g., SPI v2 confirmation fields), what's missing, and who owns it.
4. **Identify non-functional requirements** — latency (sync vs async), SLA, data volume, compliance/audit needs (relevant given your GDPR/CCPA background).
5. **Surface edge cases and failure modes early** — partial bank confirmation, late AMQ events, retries.
6. **Estimate complexity and flag risks/dependencies** before committing to a timeline.

What documents do you prepare before development starts?

- **High-Level Design (HLD):** system context diagram, sequence diagrams for the main and failure flows, integration points.
- **Low-Level Design (LLD):** class/module breakdown, DB schema changes, API contract (REST/GraphQL schema or AMQ event schema).

- **API/Event contract doc:** request/response shapes, error codes, versioning strategy (you mention SPI v2 — version negotiation is itself a deliverable).
- **Confluence technical spec** — you already do this per your resume; mention it explicitly, it shows documentation discipline.
- **Test plan:** unit, integration, and Postman-based contract tests.
- **Rollout/runbook:** feature flags, monitoring dashboards (Grafana/Datadog), rollback plan.

How do you ensure the requirement is complete and ready for development ("Definition of Ready")?

A requirement is ready when:

- Acceptance criteria are written and testable.
- API contracts are agreed upon by all consuming/producing teams.
- Dependencies (other teams, data availability) are identified and not blocking.
- Non-functional requirements (performance, security, compliance) are explicit.
- The story is sized small enough to complete within a sprint. If any of these are missing, you push back in refinement rather than letting ambiguity flow into development — this is the single most "tech lead" thing you can say.

How do you break requirements into user stories and tasks?

- Start from the **user/business outcome**, not the technical layer.
- Vertical-slice where possible: a story should deliver an observable behavior end-to-end (e.g., "merchant with unconfirmed bank account is blocked from provisioning and shown the correct contingency UI") rather than "build the DAO layer."
- Split by **risk and dependency** — do the riskiest/most uncertain piece first (e.g., the JSON Logic decision graph) so unknowns surface early.
- Break tasks within a story along technical seams: API contract, persistence, business rule logic, event consumer, tests, monitoring instrumentation.
- Keep stories small enough to demo independently — this also makes mentoring junior devs easier since each story is a complete learning unit.

How do you guide junior developers?

- Pair on the first instance of a new pattern, then let them own similar work independently.
- Review PRs for *design* issues, not just syntax — ask "why" before suggesting "what."
- Use real examples from your work — e.g., walking a junior through how the TrackMate scheduler-driven anomaly detection service was built, including what went wrong initially.

- Encourage writing tests and reading Grafana/Datadog dashboards themselves so they build an operational mindset early, not just a coding one.
 - Give them ownership of a self-contained module to build confidence (you mention mentoring 4+ engineers — be ready to describe one concrete growth story).
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3. AI Tools in Development

Do you use AI tools like GitHub Copilot, ChatGPT, or Claude?

Yes — be direct and specific since it's on your resume: Claude (Anthropic), GitHub Copilot, Cursor, and ChatGPT Enterprise. Use cases:

- Boilerplate generation (DTOs, mappers, test scaffolding).
- Exploring alternative implementations of an algorithm before committing.
- Generating first-draft documentation/Confluence specs that you then edit.
- Debugging — pasting a stack trace or log snippet to narrow down root cause faster.

How do you use AI safely in software development?

- **Never paste sensitive data** (PII, credentials, internal-only schema details) into external tools without going through approved/enterprise channels (you've used ChatGPT *Enterprise* specifically — mention that this matters because of data governance).
- Treat AI output as a **draft from a junior engineer**, not a finished artifact — it still goes through code review and tests.
- Use AI for **acceleration of known patterns**, not for **inventing business logic** you can't verify (e.g., compliance rules, risk thresholds) — those require domain validation.
- Keep a human accountable for every commit; AI suggestions don't bypass PR review or SonarQube checks.

How do you validate AI-generated code?

1. **Read it line by line** before accepting — don't blind-accept.
2. **Run existing unit/integration tests** (JUnit/Mockito) plus add new tests for the AI-suggested logic, especially edge cases.
3. **Static analysis** — SonarQube for code smells, security hotspots, complexity.
4. **Check for hallucinated APIs/libraries** — verify imports and method signatures actually exist in the version you're using.
5. **Performance/architecture sanity check** — AI tends to produce locally-correct but architecturally inconsistent code (e.g., ignoring your existing caching or exception-handling conventions); you reconcile that manually.
6. **Security review** — check for injection risks, missing input validation, especially in REST/GraphQL resolvers.

4. Problem Solving / Critical Issue

"Tell me about a critical problem you solved recently"

Pick a concrete TrackMate or Open Banking story. Example structure (STAR):

- **Situation:** TrackMate needed to classify 4,465+ web trackers across 125+ domains for GDPR/CCPA audits, and manual classification couldn't scale.
- **Task:** Reduce manual review effort while maintaining classification accuracy for legal/audit teams.
- **Action:** You designed Common Cookies grouping, Regex-based Grouped Cookies, and a scheduler-driven anomaly detection service to flag trackers needing review automatically.
- **Result:** Reduced manual review effort and improved SLA adherence for tracker classification, supporting 15 automated compliance reports covering 944K+ pages.

Have a second story ready from Open Banking (e.g., handling out-of-order or delayed AMQ Financial Instrument events causing incorrect provisioning decisions, and how you'd design idempotent/event-ordering handling).

Challenges migrating a legacy application to Spring Boot and Microservices

- **Decomposition risk:** identifying correct service boundaries without fully understanding all hidden coupling in the legacy monolith (shared DB tables, static utility classes).
- **Data migration:** splitting a shared schema into per-service schemas without breaking referential integrity; often requires a transitional shared-DB phase.
- **Distributed transactions:** legacy code likely relied on local ACID transactions; microservices need Saga or eventual consistency instead (ties into section 7).
- **Cross-cutting concerns:** auth, logging, exception handling that used to be centralized now must be reimplemented per service (or via a gateway/sidecar).
- **Testing complexity:** integration tests that used to hit one in-process call now span network calls — needs contract testing (Pact-style) and resilience testing.
- **Operational maturity gap:** legacy app likely had simple logs; microservices need distributed tracing, correlation IDs, and dashboards (Grafana/Datadog) from day one or debugging becomes very hard.
- **Team/org alignment:** services should map to team ownership boundaries (Conway's Law) — pure technical splitting without this causes coordination overhead.

Issues migrating from Oracle to PostgreSQL

- **SQL dialect differences:** Oracle's `(NVL)`, `(DECODE)`, sequences via `(NEXTVAL)`,

hierarchical queries (`CONNECT BY`) all need rewriting for PostgreSQL equivalents (`COALESCE`, `CASE`, `SERIAL`/`IDENTITY`, recursive CTEs).

- **PL/SQL to PL/pgSQL:** stored procedures and triggers (you have direct experience here) often need substantial rewriting, not a 1:1 translation.
 - **Data type mapping:** `NUMBER`, `VARCHAR2`, `CLOB`/`BLOB` differ from PostgreSQL types; precision/scale mismatches can silently corrupt data if not validated.
 - **Case sensitivity:** Oracle identifiers are typically uppercase by default; PostgreSQL is case-sensitive and lowercase by default — a common source of broken queries post-migration.
 - **Transaction/locking semantics:** differences in isolation level defaults and locking behavior can change concurrency behavior under load.
 - **Performance tuning differs:** Oracle's optimizer hints and execution plans don't map directly; indexes and query plans need to be re-tuned (`EXPLAIN ANALYZE` in Postgres).
 - **Tooling/connection pooling:** JDBC driver behavior, connection pool sizing (e.g., HikariCP defaults), and sequence/auto-increment handling need re-validation.
 - **Migration strategy:** usually dual-write or CDC-based replication (e.g., using a tool like AWS DMS) with a cutover window, plus extensive row-count/checksum validation.
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5. Microservices Architecture

How did you decide how many microservices to create / criteria to split a monolith?

Use **business capability boundaries**, not technical layers:

- **Single Responsibility at the domain level** — e.g., in TrackMate: scanning, inventory, categorization, and compliance-reporting are separate bounded contexts, each with distinct data ownership.
- **Data ownership** — if a capability owns its own tables and rarely needs joins outside them, it's a good service boundary.
- **Independent rate of change** — capabilities that change/deploy on different cadences should be separate (e.g., the risk-decisioning rules engine changes more often than core onboarding data model).
- **Team ownership (Conway's Law)** — service boundaries should roughly map to team boundaries to avoid constant cross-team coordination for every change.
- **Avoid over-fragmentation** — too many tiny services increase network hops, operational overhead, and distributed-transaction complexity; start coarser and split further only when a clear scaling or ownership pain point appears.

How do microservices communicate with each other?

- **Synchronous:** REST, GraphQL, gRPC — for request/response flows where the caller needs an immediate answer.

- **Asynchronous:** Kafka/AMQ messaging/events — for decoupled, eventually-consistent flows (your Financial Instrument AMQ events are a direct example).
- **Service mesh/gateway:** API Gateway for routing, auth, rate limiting in front of synchronous calls.

Difference between RestTemplate, Feign Client, and WebClient

	RestTemplate	Feign Client	WebClient
Style	Synchronous, blocking	Synchronous, blocking (declarative)	Reactive, non-blocking
Usage	Imperative — manually build <code>HttpEntity</code> /URLs	Declarative — interface + annotations, Spring auto-generates implementation	Functional/reactive API, supports <code>Mono</code> / <code>Flux</code>
Status	Deprecated since Spring 5 (maintenance mode)	Actively used, very common in Spring Cloud microservices	Recommended for new reactive/high-throughput code
Boilerplate	High	Low (cleanest for simple service-to-service calls)	Moderate, but enables backpressure and streaming
Best for	Legacy code maintenance only	Simple synchronous inter-service calls in a Spring Cloud ecosystem	High-concurrency, streaming, or non-blocking I/O needs

In practice: **Feign for clean, low-boilerplate synchronous service calls; WebClient when you need reactive/non-blocking behavior or are calling many downstream services concurrently; avoid RestTemplate in new code.**

When do you use synchronous vs asynchronous communication?

- **Synchronous** when the caller needs an immediate result to proceed (e.g., "is this merchant's bank confirmed *right now*" for a UI decision).
- **Asynchronous** when the result can be eventually consistent, when you want to decouple producer/consumer scaling, or when one event may trigger multiple independent downstream actions (e.g., a Financial Instrument event triggering risk re-evaluation, audit logging, and notification simultaneously).

When would you use Kafka or RabbitMQ?

- **Kafka:** high-throughput event streaming, need for replay/retention, multiple independent consumers reading the same event stream, ordered processing per partition key (good fit for your AMQ/Financial-Instrument-event use case at scale).

- **RabbitMQ**: simpler point-to-point or task-queue style messaging, lower latency for smaller message volumes, more flexible routing (exchanges/queues) for complex routing logic, easier operational footprint for moderate scale. Rule of thumb: **Kafka for event streaming/log-style architecture and high volume; RabbitMQ for traditional task/command queues.**

Why did your project use Feign Client instead of an event-driven architecture (for certain calls)?

Frame this as a deliberate trade-off, not "we just used what we had":

- Some calls in the onboarding flow need an **immediate decision** (block/allow provisioning) — that's inherently synchronous, so Feign to the risk decisioning gateway is correct there.
 - Event-driven (AMQ) is used specifically for the **asynchronous, eventually-consistent** parts — Financial Instrument confirmation events that update state independently of the user's live session.
 - Using events for everything would add unnecessary latency/complexity to flows that need a same-request answer; using synchronous calls for everything would tightly couple services and block on slow downstream events. The architecture mixes both deliberately based on each call's consistency/latency requirement.
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6. Resilience & Fault Tolerance

If one microservice goes down, how do you prevent the entire flow from failing?

- **Circuit Breaker** (Resilience4j/Hystrix-style) to stop hammering a failing dependency.
- **Fallback responses** — cached/default data or a degraded-but-functional response instead of a hard failure.
- **Timeouts** on every outbound call so one slow service doesn't exhaust threads/connections.
- **Bulkheads** — isolate thread pools/connection pools per dependency so one failing service can't starve resources needed by others.
- **Retries with backoff** for transient failures (not for non-idempotent operations without care).
- **Async/queue-based decoupling** where possible, so a downstream outage doesn't block the upstream caller at all.

Explain Circuit Breaker pattern

A Circuit Breaker wraps a remote call and tracks failure rate. If failures exceed a threshold, it "opens" the circuit and stops calling the dependency for a cooldown period, immediately returning a fallback instead — protecting the system from cascading failures and giving the failing dependency time to recover.

States of a Circuit Breaker

1. **Closed** — calls pass through normally; failures are counted.
2. **Open** — failure threshold exceeded; calls are short-circuited immediately to a fallback without hitting the dependency.
3. **Half-Open** — after a cooldown, a limited number of trial calls are let through to test if the dependency has recovered. If they succeed, the circuit closes; if they fail, it reopens.

What threshold values would you configure?

Typical starting points (then tune with real traffic data):

- **Failure rate threshold:** 50% over a sliding window (e.g., last 20–50 calls).
- **Wait duration in open state:** 5–30 seconds depending on the dependency's typical recovery time.
- **Sliding window size:** 10–100 calls or a time-based window (e.g., 60s), depending on traffic volume.
- **Slow-call duration threshold:** define what counts as "slow" (e.g., >1s) and a slow-call rate threshold, since latency degradation often precedes outright failure.
- For something like a risk decisioning call in Open Banking, you'd tune these conservatively (smaller window, faster trip) since correctness/speed matters for onboarding UX.

How do Retry and Fallback methods work?

- **Retry:** re-attempts a failed call a configured number of times, usually with exponential backoff and jitter, only for transient/idempotent-safe failures (timeouts, 5xx) — never blindly retry non-idempotent writes without deduplication keys.
 - **Fallback:** defines an alternative response path when the primary call fails or the circuit is open — e.g., return cached bank-confirmation status, a default "pending" decision, or a degraded UI state — so the caller gets *something* usable instead of an exception bubbling up.
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7. Data Consistency in Microservices

How do you maintain data consistency across multiple microservices?

Since each service owns its own database, you can't rely on a single ACID transaction.

Approaches:

- **Saga pattern** for multi-step business transactions across services (see below).

- **Eventual consistency** via events — each service updates its own state and publishes an event; other services react asynchronously.
- **Idempotency keys** on writes/event consumers so retries or duplicate events don't corrupt state.
- **Outbox pattern** — write the event to an "outbox" table in the same local transaction as the business write, then a separate process publishes it, guaranteeing the event isn't lost even if publishing fails.

Explain Saga pattern

A Saga breaks a distributed transaction into a sequence of local transactions, each in a different service, with **compensating actions** defined for each step. If a later step fails, previously completed steps are rolled back via their compensating transactions (rather than a single atomic rollback, which isn't possible across services).

Orchestration vs Choreography Saga

	Orchestration	Choreography
Control	A central orchestrator service explicitly tells each service what to do next	No central controller; each service reacts to events and publishes its own events
Visibility	Easier to trace/debug — one place to see the whole flow	Harder to trace — flow logic is distributed across services
Coupling	Orchestrator is coupled to all participants	Services are decoupled from each other, only coupled to event contracts
Best for	Complex, multi-step business processes needing clear control and easy monitoring	Simple flows with a small number of participants where loose coupling matters more

Have you implemented Saga in a real project?

If you haven't formally implemented a named "Saga" framework, be honest but bridge to what you *have* done: your asynchronous AMQ-driven decisioning in Open Banking (provisioning gated on a separate bank-confirmation event, with state updates flowing independently) is conceptually a choreography-style eventual-consistency flow. You can say: "I haven't used a formal Saga orchestration framework, but I've built the underlying pattern — async event consumption that updates state independently and needs to handle out-of-order or failed events, which is the same consistency problem Saga solves. I'd be comfortable formalizing this with compensating transactions if the use case required strict rollback semantics."

What is CQRS?

Command Query Responsibility Segregation — separating the **write model** (commands that change state) from the **read model** (queries that read state), often backed by different data stores/schemas optimized for each purpose.

When should CQRS be used?

- When read and write workloads have very different scaling or modeling needs (e.g., complex write-side validation/business rules vs. simple, high-volume read-side reporting — relevant to your TrackMate compliance reporting, which reads aggregated data very differently from how trackers are written/classified).
- When you need to scale reads and writes independently (e.g., heavy reporting load shouldn't impact transactional write performance).
- When combined with event sourcing for full audit trails (useful in compliance-heavy domains like yours).
- **Not** needed for simple CRUD services — it adds real complexity and should be reserved for genuinely asymmetric read/write needs.

Operational challenges of CQRS

- **Eventual consistency between read and write models** — the read side may lag behind the write side, which needs to be communicated to users/UI.
 - **Increased infrastructure complexity** — maintaining separate data stores/schemas, sync pipelines between them.
 - **Debugging complexity** — a bug could be in the write model, the sync mechanism, or the read model.
 - **Operational overhead** — more moving parts to monitor (you'd lean on your Grafana/Datadog experience here) and more deployment surface area.
 - **Data duplication** — read models often denormalize data, increasing storage and requiring careful change-management when the write-side schema evolves.
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9. Hiring & Leadership

If you were hiring a Java developer, what would you evaluate?

- **Fundamentals over framework trivia:** OOP, SOLID, data structures — can they reason about *why* a design works, not just recite Spring annotations.
- **Debugging approach:** give them a broken piece of code/log and watch their process, not just the answer.

- **Practical Spring Boot/REST/DB experience:** can they design a simple API and schema for a given problem on the spot.
- **Communication:** can they explain a past project clearly and own trade-offs they made (mirrors what you're doing in this interview).
- **Testing discipline:** do they think about edge cases and write tests without being told to.
- **Curiosity/growth signal:** have they learned something recently beyond what their job required (relevant given your own AI-tools adoption).

What technical areas are most important for a 2-4 year developer?

- Solid Java fundamentals (collections, concurrency basics, exception handling) and Spring Boot proficiency.
- Comfortable with REST API design and basic SQL/query optimization.
- Understands testing (JUnit/Mockito) and version control workflows.
- Beginning to understand distributed systems concepts (caching, idempotency, basic resilience) even if they haven't built them from scratch yet.
- Can read and reason about existing code, not just write new code in isolation.

How do you assess debugging and problem-solving skills?

- Present a real (sanitized) bug or a failing test and ask them to walk through their thought process out loud.
- Watch whether they form a hypothesis and test it methodically, vs. randomly changing code.
- Ask how they'd investigate a production issue with only logs/dashboards (ties directly to your Grafana/Datadog/CAL-logs experience) — do they know how to use observability tools, not just a debugger.
- Ask about a past bug they're proud of fixing — depth of explanation reveals real understanding vs. memorized answers.

10. Coding Round

Question 1 — Smallest Missing Natural Number

Input: `[1, 5, 2, 6, 8, 4, 9]` → **Output:** `3`

Approach: The smallest missing natural number for an array of size n must lie between 1 and $n+1$. Use this bound to solve in $O(n)$ time and $O(1)$ extra space (in-place index marking), or $O(n)$ time/space with a boolean array (simpler to explain in an interview).

```
java
```

```

import java.util.*;

public class SmallestMissingNumber {

    // O(n) time, O(n) space – simplest to explain and code under pressure
    public static int findSmallestMissing(int[] nums) {
        int n = nums.length;
        boolean[] present = new boolean[n + 2]; // indices 1..n+1

        for (int num : nums) {
            if (num >= 1 && num <= n + 1) {
                present[num] = true;
            }
        }

        for (int i = 1; i <= n + 1; i++) {
            if (!present[i]) {
                return i;
            }
        }
        return n + 1; // unreachable given the boolean array size, but safe fall
    }

    // O(n) time, O(1) extra space – optimal version (in-place placement / Dutch
    public static int findSmallestMissingOptimal(int[] nums) {
        int n = nums.length;
        for (int i = 0; i < n; i++) {
            // place each value v (1..n) at index v-1, if possible
            while (nums[i] > 0 && nums[i] <= n && nums[nums[i] - 1] != nums[i]) {
                int correctIdx = nums[i] - 1;
                int temp = nums[correctIdx];
                nums[correctIdx] = nums[i];
                nums[i] = temp;
            }
        }
        for (int i = 0; i < n; i++) {
            if (nums[i] != i + 1) {
                return i + 1;
            }
        }
        return n + 1;
    }

    public static void main(String[] args) {
        int[] input = {1, 5, 2, 6, 8, 4, 9};
        System.out.println(findSmallestMissing(input.clone())); // 3
    }
}

```

```
        System.out.println(findSmallestMissingOptimal(input.clone())); // 3
    }
}
```

Talking points for the interviewer: explain the bound (answer is always within $[1, n+1]$), discuss why a `HashSet` approach also works ($O(n)$ time/space, easier to write but uses more memory than the in-place version), and mention the in-place version as the "optimal" follow-up if they push for $O(1)$ space.

Question 2 — Longest Common Substring

Input: `S1 = "Hare Krishna"`, `S2 = "Mister Krishna Yadav"` → **Output:** `"Krishna"`

Approach: Classic Dynamic Programming — build a table `dp[i][j]` representing the length of the common substring ending at `S1[i-1]` and `S2[j-1]`. Track the maximum length and its ending position to reconstruct the substring. Time: $O(m \cdot n)$, Space: $O(m \cdot n)$ (can be optimized to $O(n)$ with rolling arrays).

java

```

public class LongestCommonSubstring {

    public static String longestCommonSubstring(String s1, String s2) {
        int m = s1.length(), n = s2.length();
        int[][] dp = new int[m + 1][n + 1];
        int maxLen = 0;
        int endIndexInS1 = 0; // exclusive end index in s1 of the best match

        for (int i = 1; i <= m; i++) {
            for (int j = 1; j <= n; j++) {
                if (s1.charAt(i - 1) == s2.charAt(j - 1)) {
                    dp[i][j] = dp[i - 1][j - 1] + 1;
                    if (dp[i][j] > maxLen) {
                        maxLen = dp[i][j];
                        endIndexInS1 = i;
                    }
                } else {
                    dp[i][j] = 0; // substring (contiguous) breaks here
                }
            }
        }
        return s1.substring(endIndexInS1 - maxLen, endIndexInS1);
    }

    public static void main(String[] args) {
        String s1 = "Hare Krishna";
        String s2 = "Mister Krishna Yadav";
        System.out.println(longestCommonSubstring(s1, s2)); // Krishna
    }
}

```

Talking points: clarify *substring* (contiguous) vs *subsequence* (not necessarily contiguous) up front — interviewers often probe this distinction. Mention the space optimization (only need the previous DP row, reducing space to $O(n)$) as a follow-up if asked to optimize.

11. Performance Optimization

How do you optimize a Spring Boot application?

- **Connection pooling** tuned correctly (HikariCP pool size matched to DB capacity and thread count).
- **Caching** frequently-read, rarely-changed data (Redis — which is on your stack) to cut DB round trips.
- **Lazy vs eager loading** tuned deliberately in JPA relationships to avoid over-fetching.

- **Async/non-blocking** where appropriate (`@Async`, WebClient, or reactive stacks) to free up threads.
- **JVM tuning**: appropriate heap sizing, GC algorithm choice (e.g., G1/ZGC) based on latency requirements.
- **Profiling before optimizing** — use Datadog/Grafana/Prometheus metrics (your actual stack) to find the real bottleneck rather than guessing.
- **Pagination and projections** instead of fetching full entities/large result sets.

What is the N+1 query problem?

When fetching a list of parent entities (1 query) and then, for each parent, lazily fetching a related child collection (N additional queries) — resulting in N+1 total queries instead of one efficient query. This typically shows up with JPA/Hibernate lazy associations accessed inside a loop.

How do you solve N+1 queries in JPA/Hibernate?

- `JOIN FETCH` in JPQL to eagerly fetch the association in the same query when you know you'll need it.
- `@EntityGraph` to declaratively specify which associations to fetch eagerly for a given query, without changing the global fetch type.
- **Batch fetching** (`@BatchSize` or `hibernate.default_batch_fetch_size`) to fetch related entities in batches instead of one-by-one.
- **DTO projections** — query only the fields you need via constructor expressions or Spring Data projections, avoiding entity graph traversal entirely for read-heavy endpoints.
- **Monitor with SQL logging/Datadog APM** to actually catch N+1 patterns in practice, since they're easy to introduce unintentionally.

How do caching and static data improve performance?

- Caching avoids repeated expensive computation or DB/network round trips for data that doesn't change often (e.g., reference/lookup data, classification rules in TrackMate-style systems).
- **Static/reference data** (e.g., common cookie categories, regex group definitions) can be loaded into memory or Redis at startup/refresh intervals rather than queried per request, dramatically cutting latency for high-volume read paths.
- Multi-tier caching (which you've explicitly built — "30+ GraphQL APIs with multi-tier caching") balances freshness vs. speed by layering in-memory (fast, smaller) and distributed (Redis, larger, shared across instances) caches.

How do you optimize database queries?

- **Indexing** appropriately on filter/join/sort columns (and avoiding over-indexing, which slows writes).

- **Query plan analysis** (`EXPLAIN`/`EXPLAIN ANALYZE`) to find full table scans or missing index usage.
- **Avoiding** `SELECT *` — fetch only needed columns.
- **Batching writes** instead of row-by-row inserts/updates.
- **Using CTEs and proper joins** (on your resume) instead of correlated subqueries where possible, for readability and often better optimizer behavior.
- **Aggregate/materialized tables** for expensive reporting queries (you built "aggregate tables" for TrackMate reporting — a great concrete example here).
- **Stored procedures/triggers judiciously** — useful for enforcing data integrity close to the data, but be ready to discuss when application-layer logic is preferable (testability, version control, portability).

How do utility classes and design patterns help performance and maintainability?

- **Utility classes** centralize repeated logic (validation, formatting, mapping) so it's optimized and tested once, not reimplemented per-module — reduces bugs and improves consistency.
- **Design patterns** (Strategy, Factory, Builder, Decorator, Singleton where appropriate) make code easier to extend without modification (Open/Closed Principle), which reduces regression risk and speeds up future changes — directly relevant to a SOLID/design-patterns-heavy resume like yours.
- Example from your own experience: **MapStruct** for object mapping avoids hand-written, error-prone, and slower reflection-based mapping; **Regex-based Grouped Cookies** is itself a pattern abstraction (rule-based grouping rather than hardcoded per-cookie logic) that scales better as the inventory grows (4,465+ trackers).

Quick Reference — "Most Important to Master" Checklist

- ☐ Requirement analysis & HLD — practice a 2-minute walkthrough using Open Banking as the example.
- ☐ Monolith → microservices — know the boundary criteria (data ownership, rate of change, team ownership) cold.
- ☐ Feign vs WebClient vs Kafka — know the table above by heart and **why** each was chosen in your project.
- ☐ Circuit Breaker + Retry + Fallback — states (Closed/Open/Half-Open) and realistic threshold numbers.
- ☐ Saga pattern — Orchestration vs Choreography table, and your honest "I haven't formally implemented it but here's the equivalent pattern I have built" answer.
- ☐ CQRS — when to use it, and at least one operational challenge to mention unprompted.
- ☐ N+1 query optimization — `JOIN FETCH`, `@EntityGraph`, batch fetching, DTO projections.

- ☐ Smallest missing number — practice typing the boolean-array version cold, then mention the $O(1)$ -space optimal version.
- ☐ Longest common substring — practice the DP table version, and be ready to explain substring vs. subsequence.
- ☐ Leadership & hiring approach — have one concrete mentoring story (a specific engineer/situation) ready, not a generic answer.

Good luck — your resume already gives you real, specific stories for almost every question above. The main job in the interview is retrieval speed: practice saying these out loud once or twice so the right project example comes to mind instantly under pressure.